Operating Systems and System Programming

Fall 2026

Class overview

1. **Grading:**
   - Attendance : 30
   - Assignments : 20
   - Midterm : 20 (In-person)
   - Final : 30 (In-person)

2. **Platforms:**
   - For class materials: **LMS**
   - For communication: **LMS** and **Kakao Chatroom**

3. **Contacts**:
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Kakao QR Code



Supporting materials:

- Systems programming and operating systems, Dhamdhere, D. M (1993)
- Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2018). *Operating Systems: Three Easy Pieces*. Arpaci-Dusseau Books. (*Retrieved from https://pages.cs.wisc.edu/~remzi/OSTEP/)*
- Wienand, I. (2007). *Computer Science from the Bottom Up*. (Retrieved from http://www.bottomupcs.com/)

Chapter I:
Introduction to Operating System and System Programming

I.1. Overview

This chapter introduces generalities about system operating systems and system programming. It uncover important concepts and discusses about applications.

This introductory chapter lays foundations that will help students gain practical insight on the subject.

Outline

- What is an Operating System (OS)?
- Type of Operating Systems
- Why Python for OS Exploration?
- What is System Programming (SP)?
- Key areas in SP
- System Programming in Python
- Conclusion

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- **What is an Operating System (OS)?**
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I.2. What is an Operating System (OS)?

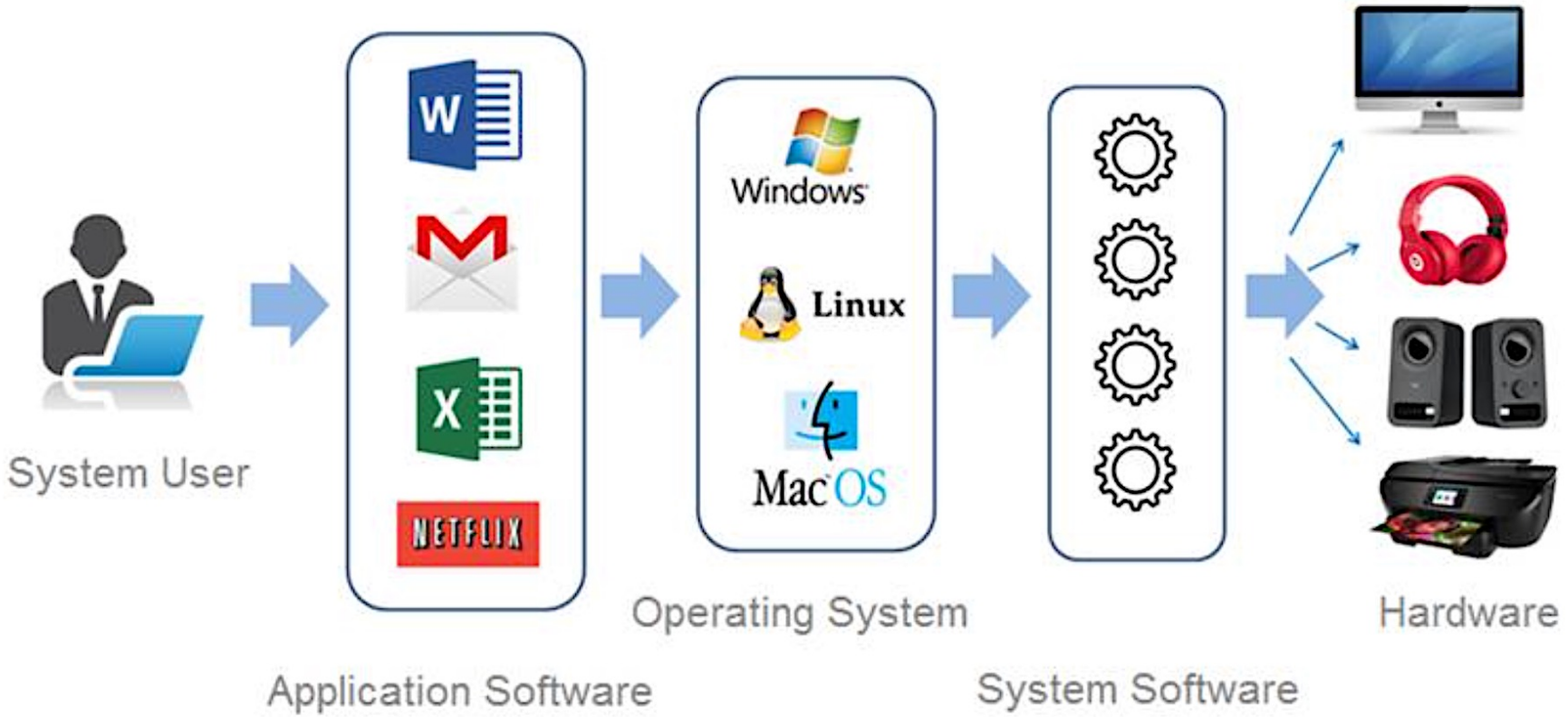
An Operating System (OS) is a software program that:

- ✓ manages computer hardware, software resources, and
- ✓ provides common services for computer programs.

It acts as an *intermediary* between users and the computer hardware.

System software refers to a category of computer program designed to operate, control and extend the capabilities of the computer hardware.




System User
Windows
Linux
Mac OS
NETFLIX
Application Software
Operating System
System Software
Hardware

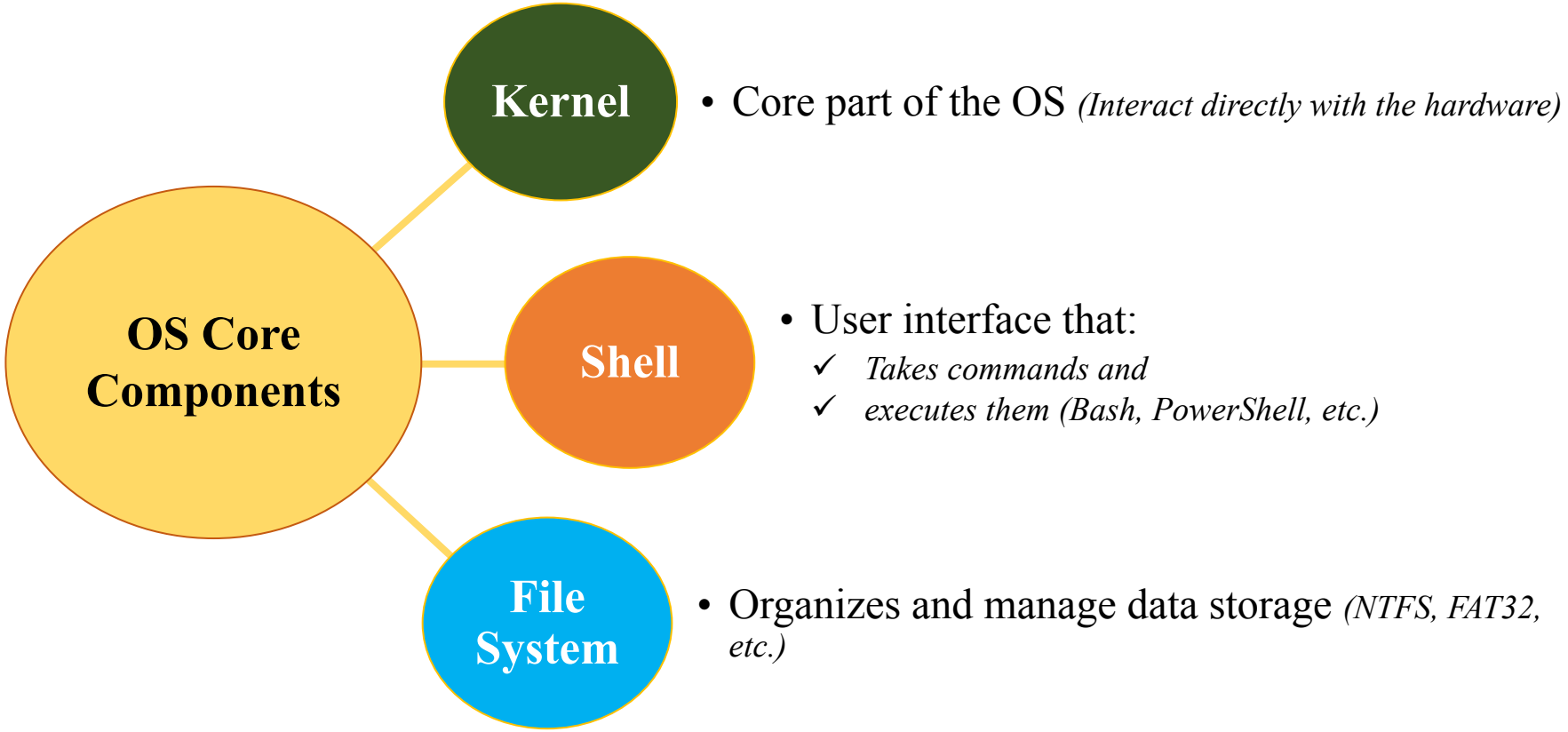
I.2.1. Main role of OS

OS plays critical role in:

- **Process Management**: Manages the execution of multiple processes.
- **Memory Management**: Allocates and deallocates memory for programs.
- **File System Management**: Handles file storage, retrieval, and permissions.
- **Device Management**: Controls and coordinates hardware components.
- **User Interface**: Provides a way for users to interact with the system (GUI or CLI).

I.2.2. Core components of an OS

The followings are the key components of any OS:



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I.3. Type of Operating System

Operating systems can be classified based on their *design, functionality, and usage*.

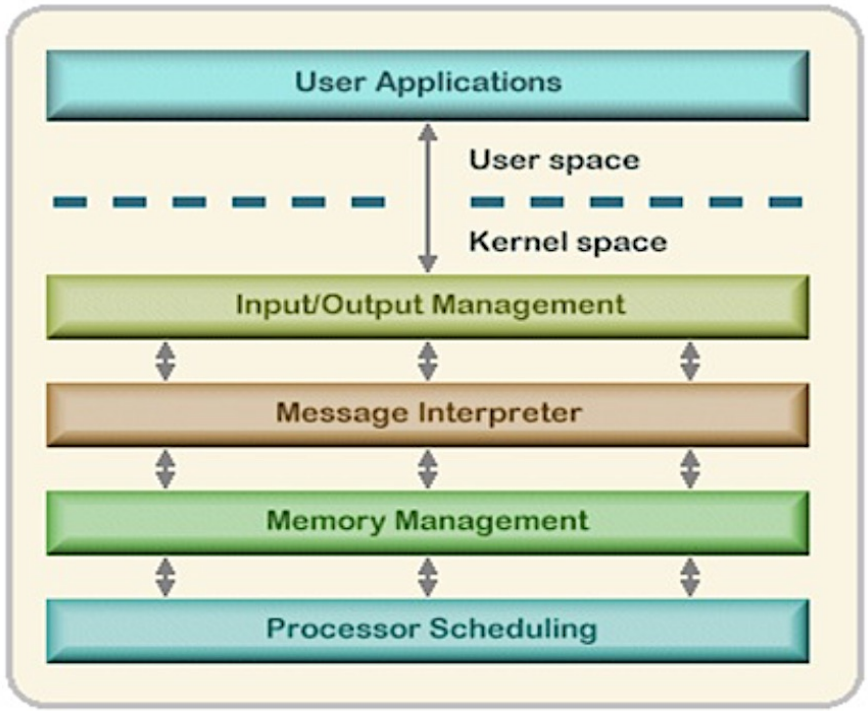
Thus, we have:

Type	Description	Examples
Single-User OS	Support one user at a time, with limited multitask	MS-DOS, windows 95
Multi-User OS	Allows multiple users to access the system simultaneously	Linux, Unix, Windows server
Batch OS	Executes jobs in batches without user interaction	IBM OS
Real-time OS	Processes data in real-time with strict timing constraints	VxWorks, QNX
Distributed OS	Manages a group of networked computers as a single system	Amoeba, Plan9
Embedded OS	Designed for embedded systems with limited resources	Embedded Linux
Mobile OS	Optimized for smartphones and tablets	Android, IOS

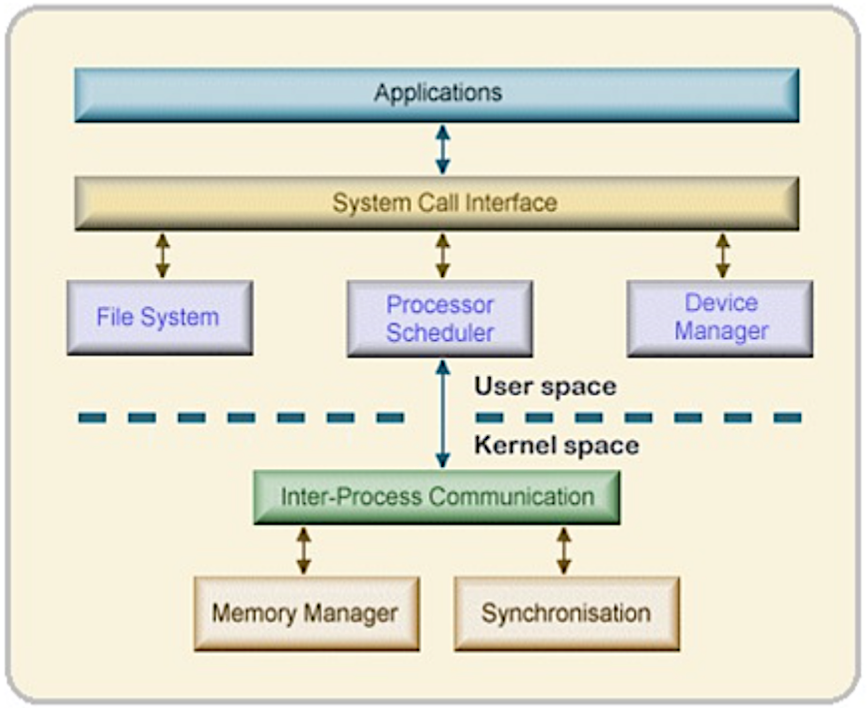
These OS presents some similarities and differences that can be summarized as follows:

Aspects	Similarities	Differences
Kernel	All OS have a kernel for hardware control	Monolithic (Linux) vs Microkernel (QNX)
User Interface	Most provide GUI & CLI	Some (e.g., servers) are CLI-only
Multitasking	Modern OS support multitasking	Real-time OS prioritize task scheduling
Security	All enforce user permissions	Multi-user OS have stricter access control

Monolithic



Microkernel



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I.4. Why Python for Operating System Exploration?

Python is widely used for OS interaction due to its *simplicity* and *rich libraries*, while C is traditionally used for low-level OS development

Features	Python	C
Ease of use	High-level, easy to learn and write	Low-level, complex memory management
Speed	Slower (interpreted)	Faster (compiled, direct hardware access)
Libraries	Rich ecosystem	Limited high-level abstractions
Portability	Cross-platform (works on Windows, Linux)	Requires recompilation for different OS
Use case	Scripting, automation, rapid prototyping	Kernel development, system programming

Good

Acceptable

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I.5. What is System Programming (SP)?

It refers to the development of software that:

- ❑ provides core services to the computer hardware and
- ❑ enables the execution of application programs.

It involves writing *low-level code that interacts directly with the operating system (OS), hardware, and system resources.*

I.5.1. Main characteristics of System Programming (SP)?

SP is:

- **Close to Hardware**: Deals with memory management, process control, and device drivers.
- **Interacts with OS**: Uses system calls to request services from the kernel.
- **Performance-Critical**: Requires efficient resource management (CPU, memory, I/O).
- **Written in Low/Medium-Level languages**: Primarily C, C++, Rust, and sometimes Python (for scripting).

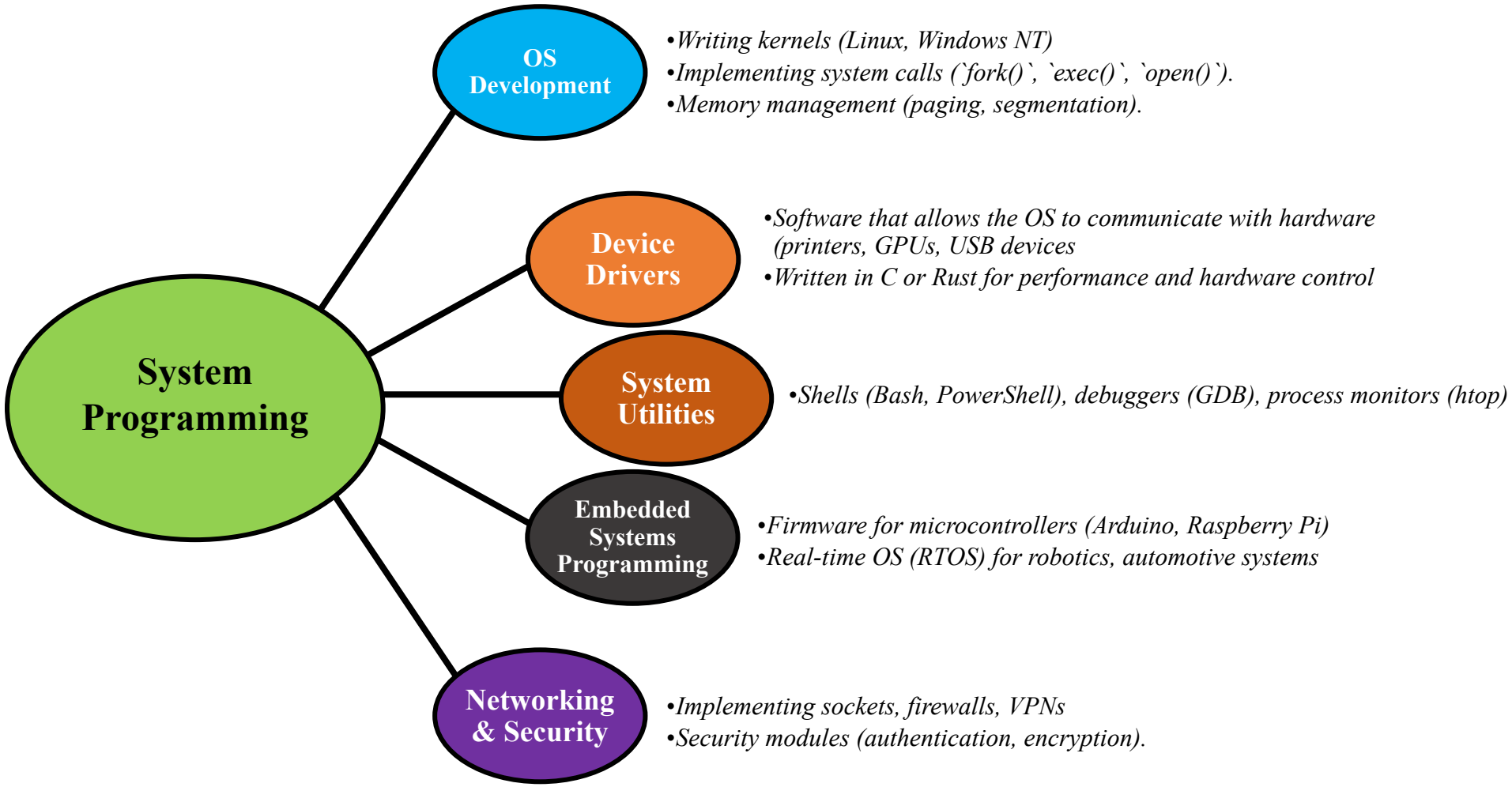
I.5.2. Difference between System Programming and Application Programming

Aspects	System Programming	Application Programming
Level of abstraction	Low-level (direct hardware/OS interaction)	High-level (user-facing software)
Languages used	C, C++, Assembly, Rust	Python, Java, JavaScript, C#.
Purpose	Manages hardware & OS resources	Solves end-user problems (web apps)
Examples	OS kernels, device drivers, shells	Web browsers, games, office software

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I.5.3. Key areas of System Programming (SP)



Thus, SP helps:

- ❑ Understand how computers work: Learning CPU, memory, and I/O management.
- ❑ Build efficient software: Optimizing performance-critical applications.
- ❑ Work on OS/Embedded Systems: Careers in kernel development, IoT, cybersecurity.
- ❑ Debug complex issues: Diagnosing low-level errors in applications.

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I.6. Learning SP with Python vs C

Rationales behind using Python rather than C are summarized below:

Feature	Python (for System Scripting)	C (Traditional System Programming)
Level of abstraction	High-level (easy to use)	Low-level (direct hardware access)
Performance	Slower (interpreted)	Faster (compiled, optimized)
Use case	Automation, scripting, rapid prototyping	Kernel/driver development, real-time systems
Examples	‘psutil’ for process monitoring	Writing a Linux device driver

Good

Acceptable

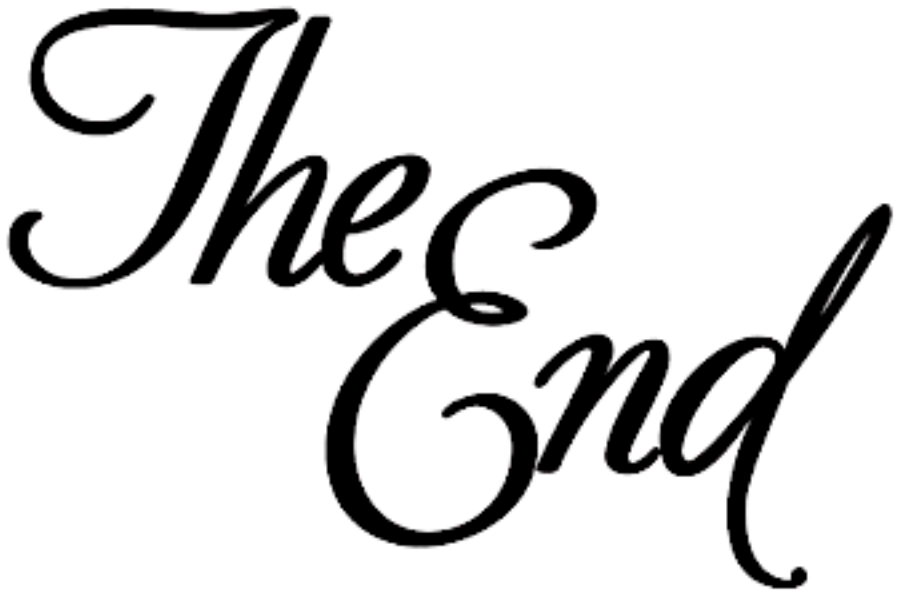
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I.7. Conclusion

OS and SP are and remain fundamental notions in computer science. In this chapter core information about them, and thoroughly provided with clear explanation regarding their their unique relevance. Having this paves the way to deeper understanding and learning of the topic. It opening the way for further investigations and deeper knowledge.

Note:
Images source: (https://www.learncomputerscienceonline.com/operating-system/)


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